

LETTER

A 12 μ A –82 dB PSR LDO with PSR enhance technique and dynamic compensation technique

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Abstract This article presents a novel low dropout regulator with high PSR and reliable stability. In order to achieve high PSR, this design use PSR enhance circuit to filter the voltage ripple from reference at low frequency, and achieving –82 dB PSR at 1 KHz with the load of 20 mA. Besides, owing to the dynamic RC compensation network, this design can reconfigure the compensation resistor and capacitor, and achieve good stability at full load up to 250 mA. Finally, this design was implemented with the 0.5- μ m CMOS technology, and the total quiescent current of the proposed LDO is only 12 μ A.

Keywords: low dropout regulator (LDO), low quiescent current, high power supply rejection (PSR), dynamic RC compensation

Classification: Integrated circuit

1. Introduction

In recent years, portable devices such as smartphones and wearable electronics have gained significant popularity due to their compact size and lightweight design [1–4]. These devices typically integrate multiple chip modules, including comparators, amplifiers, switches, and analog to digital converters, all of them impose stringent demands on the DC and AC performance of the power supply module. The commonly adopted power solution involves the DC-DC converter and low dropout regulator (LDO). The DC-DC converter generally has an output ripple voltage about 20 mV to 50 mV which can affect the chip performance. This ripple voltage will be connected to a LDO which can suppress the amplitude of ripple voltage to below 3 mV [5, 6]. In general, 3 mV ripple voltage has minimal influence to the electronic devices. Consequently, the power supply ripple rejection of the LDO is important to the performance of the integrated chip modules [7–10].

The traditional LDO architecture is shown in Fig. 1, which primarily comprises a reference, an error amplifier, a power transistor and feedback resistors [11–14]. But in general, the high output impedance of the error amplifier combined with the considerable parasitic capacitance at the gate of the power transistor can generates a low-frequency pole. To address this issue, a buffer stage is inserted to separate the big output impedance at the output of the first stage and big

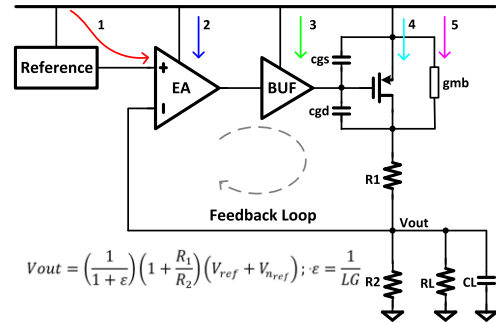


Fig. 1 The conventional LDO structure.

parasitic capacitance at the gate of power stage [15–17].

In recent years, many scholars have proposed lots of methods to mitigate power supply ripple voltage. The most popular techniques for enhancing PSR can be categorized into four types. The first approach is placing an RC filter in series with the power supply of the LDO. This technique can attenuate power supply ripple beyond the bandwidth of the RC circuit. However, when under heavy load conditions, this method can lead to a significant voltage drop which adversely affecting the load regulation of the LDO [18–20]. The second method focuses on increasing the closed loop gain and bandwidth of the LDO. In the power supply rejection curve, the suppression effect is positively correlated with the loop gain, while the suppression frequency range is directly related to the loop bandwidth. Nevertheless, excessively high loop gain poses substantial challenges to stability and can result in additional current consumption [21–23]. The third method is cascading two LDOs. This strategy can effectively enhance the power supply rejection, however, it brings relatively high hardware costs and more power consumption [24, 25]. The fourth approach is to build a feedforward path by introducing a copy signal at the gate of the power transistor, thereby stabilizing the VGS of the PMOS power transistor [26]. However, this method is relatively complex and may lack stability under process, voltage, and temperature (PVT) variations.

The proposed design focuses on improving low-frequency power supply rejection by utilizing a PSR enhancement technique to suppress ripple voltage from the reference circuit. Additionally, a fast startup path has been constructed to decrease the startup time of the LDO. Furthermore, this design also integrates the dynamic compensation technique to enhance loop stability.

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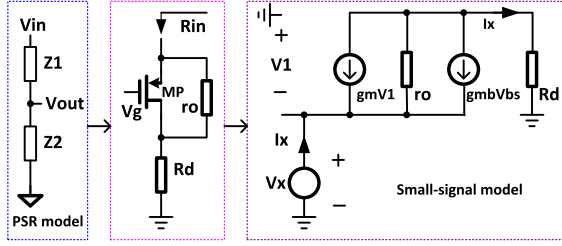


Fig. 2 The small signal model of the power stage.

2. The proposed LDO

2.1 The analysis of low frequency PSR

Researching the five sources of power supply ripples in Fig. 1, they can be categorized into two types based on whether they are modulated by the closed loop. The first type includes power supply ripple that is affected by loop regulation, including ripple sources 2, 3, 4, and 5. The second type consists of power supply noise that is not modulated by the closed loop, such as ripple source 1 in Fig. 1, which come from the reference module and is independent of the closed loop [27, 28].

In order to analyze the ripple path that are influenced by the loop, the small-signal analysis method is used as shown in Fig. 2. Comparing with the voltage divider model in Fig. 2, the impedance of the power transistor in the core loop can be equivalently represented as Z_1 , while the load resistance and feedback resistor can be equivalently represented as Z_2 [29, 30].

In Fig. 4 shown later, the input signal is applied at the source terminal of the power transistor, while the output signal is taken from the drain terminal. In order to simplify the analysis, Z_1 will not be calculated at this moment, the power transistor can be considered as a common-gate amplifier with its gate connected to a fixed voltage. According to the small-signal model of the common-gate amplifier, the input impedance R_{in} at the input terminal is given by the following equations.

$$V_1 = -V_x \quad (1)$$

$$R_d I_x + r_o (I_x - (g_m + g_{mb}) V_x) = V_x \quad (2)$$

$$\frac{V_x}{I_x} = \frac{R_d + r_o}{(1 + (g_m + g_{mb}) r_o)} \quad (3)$$

$$R_{in} = \frac{V_x}{I_x} = \frac{R_d + r_o}{(g_m + g_{mb}) r_o} = \frac{R_d}{(g_m + g_{mb}) r_o} + \frac{1}{g_m + g_{mb}} > \frac{1}{g_m + g_{mb}} \quad (4)$$

$$LG = A_{v1} A_{v2} A_{v3} \beta \gg 1 \quad (5)$$

$$R_{in_{loop}} = R_{in}(1 + LG) \quad (6)$$

$$Z_1 = R_{in_{loop}} - Z_2 = R_{in}(1 + LG) - Z_2 \quad (7)$$

$$PSR = \frac{V_{out}}{V_{in}} = 20 \lg \left(\frac{Z_2}{Z_1 + Z_2} \right) = 20 \lg \left(\frac{Z_2}{R_{in}(1 + LG)} \right) \quad (8)$$

As can be seen from the above equation, due to the feedback of closed loop, the noise voltage of path 2 in Fig. 3 will

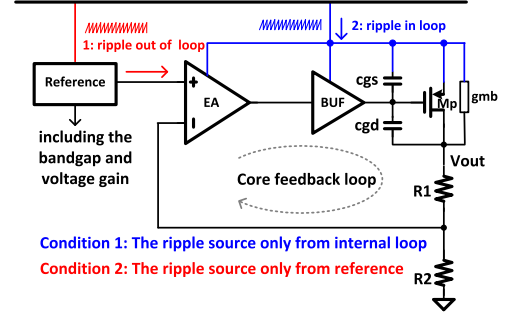


Fig. 3 The voltage ripple in the loop and out of the loop.

be attenuated by loop modulation. The power supply ripple at the output will be very small. Moreover, the greater the loop gain, the more obvious the suppression of ripple 2.

After analyzing ripple 2 in Fig. 3, then analyze ripple 1 in Fig. 3. It is clearly that ripple 1 mainly comes from the reference which including the bandgap and the voltage gain module. Traditional bandgap references do not have high power supply rejection because the limited loop gain and loop bandwidth. In general, the loop gain and bandwidth of bandgap is much smaller than LDO core loop. there will be more voltage ripple from bandgap to the LDO output than from the ripple 2. So, the bandgap will be the mainly ripple source to the output voltage of the LDO and decrease the PSR of LDO in low frequency.

2.2 The low frequency PSR enhance circuit

To further enhance the effect of low-frequency ripple suppression, this design adds a low frequency PSR enhancement circuit. In fact, this low frequency PSR enhancement module can be regarded as a RC filter. A large resistor and a large capacitor are used to achieve filtering the reference ripples in the low frequency. However, the big resistor occupies a large area and is not suitable for integrate into the chip. In this design, an active transistor is used as the filter resistor. When the transistor works in the linear region, it is equivalent to a large resistor. In addition, in order to effectively utilize the remaining area of the chip, the capacitor in the filter is also build by transistors which connect the drain terminal, source terminal and body terminal together to ground as the negative terminal, While the gate of the transistor as the positive terminal of the capacitor.

As shown in the Fig. 4, MP4 is the filter resistor in this design, and how to get an appropriate bias voltage for MP4 is a challenge. This bias voltage must ensure that the MP4 operates in the linear region, specifically that $V_{GS} < V_{TH}$. Additionally, the bias voltage needs to adapt to variations in process, voltage, and temperature (PVT) conditions, and also tracking the variations in the threshold voltage of the MP4.

In this design, in order to provide a suitable bias voltage for MP4, a special bias path is designed to make the MP4 work in linear region. On the bias path in the above Fig. 4, MN1 can be regarded as a current source and determines the current of MP2. The source voltages of MP2 and MP4 are equal, but the transistor width of them is different while the length is the same. The width of MP2 is greater than MP4. Therefore, the voltage drop on MP2 will be less than

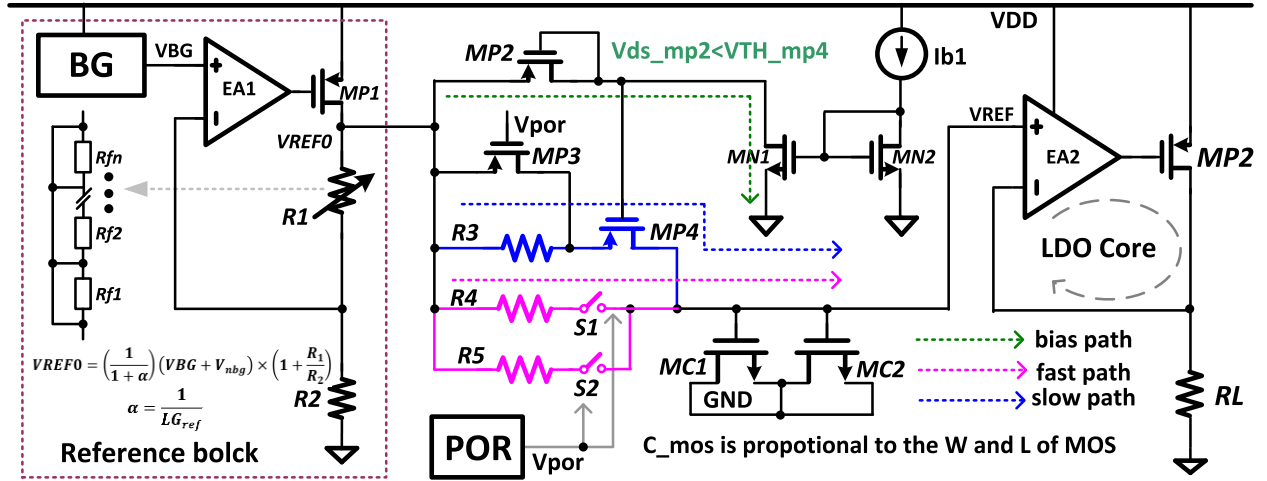


Fig. 4 The circuit of the PSR enhancement technique.

the threshold voltage of MP4, means $V_{ds_{mp2}} < V_{TH_{mp4}}$. At this time, the MP4 works in the linear region. When the size of MP2 increased larger, the equivalent resistance of MP4 will continue to increase to hundreds of M ohm or even higher. In addition, this bias voltage can follow the PVT changes of MP2. In the layout process, MP2 and MP4 need to be carefully matched to ensure that MP4 operates in the deep linear region under any process corner.

The PSR enhancement circuit is put at the slow path in Fig. 4. On the slow path, the signal will pass through resistor R3 and transistor MP4. After that, the signal will pass through the filter capacitor to the input of error amplifier. The advantage of using transistor capacitor is that the overall area is small and it only active when the gate voltage is higher than the threshold voltage.

In this design, the bandwidth frequency of the filter is about several Hz. Therefore, the power supply rejection at the 1 kHz of this design is as high as 82 dB.

2.3 The fast startup circuit

This design uses PSR enhance technique to enhance low frequency PSR, but this technique has some disadvantages. During the startup process, the VREF output node VREF0 needs to charge the VREF1 node fast. However, due to the large filter, the charging speed will be relatively slow, resulting in a long startup time. To avoid the startup issue, this design constructs a fast startup path. When the LDO is enable, VREF0 will charge the VREF1 through the fast path quickly at the first time, after that, the fast startup path will be closed and enable the PSR enhancement circuit at the slow path to decrease the ripple from reference.

The fast startup path in this design is composed of POR, R4, and R5. During the startup process, the enable signal go to the POR circuit and generates a flag signal. This signal will turn on the fast startup path. At this time, the PSR enhancement path is disabled and the resistance on the fast path is very small. Since the charging time is inversely proportional to the resistor in filter. Therefore, the smaller the resistor, the shorter the startup time.

2.4 The stability analysis of LDO loop

In traditional LDO circuits, the structure of the error ampli-

fier and power transistor is crucial and related to the loop stability and dc also ac characteristics. As shown in Fig. 5, the proposed LDO adopt a three-stage structure. The first stage is a folded cascade amplifier. Among them, MN1 and MN2 are the input transistors, and NM3 is the tail current source of the first stage. This part of current is not a fixed current and will tracking the load current changes. The second stage is a typical common-source amplifier, which can split the large impedance of the first-stage output and the large capacitance at the gate of the third-stage power transistor, and avoid generating a lower pole. The third stage is the power transistor, this design use PMOS as the power transistor, which is helpful for achieving high PSR.

In this design, a dynamically compensation network structure is adopted. the compensation resistor can automatically change its size tracking the load current size. In order to analyze the loop stability, the small signal model of the LDO is drawn and shown in Fig. 6.

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{(g_{m4} + g_{m3}k_2)(sC_L R_L + 1)(sC_c R_c + 1)g_{m1}}{R_L(sC_c R_c R_1 + R_1 - sC_c K_1)} \quad (9)$$

$$k_1 = \frac{s^2 g_{m2} C_c R_c C_2 - s C_c R_c - 1}{s^2 C_c C_2 + s C_c R_c + 1} \quad (10)$$

$$k_2 = \frac{g_{m2}(sC_c R_c + 1)(sC_2 R_2 + 1)}{s^2 C_c C_2 R_2 + s C_c R_c R_2 + R_2} \quad (11)$$

$$p_1 = \frac{C_c(g_{m2}R_2 - 1) - C_1}{C_1 C_c R_1 + C_1 C_c R_c - g_{m2}R_2 C_1 C_c R_1} \quad (12)$$

$$p_2 = \frac{1}{R_2 C_2} \quad (13)$$

$$p_3 = \frac{1}{R_L C_L} \quad (14)$$

$$z_1 = \frac{g_{m3}}{g_{m2}g_{m3}C_2 R_2 + g_{m3}C_c R_c g_{m2} + g_{m4}C_2} \quad (15)$$

$$z_2 = \frac{g_{m2}g_{m3}C_2 R_2 + g_{m3}C_c R_c g_{m2} + g_{m4}C_2}{g_{m2}g_{m3}C_2 R_2 C_c R_c} \quad (16)$$

In the above equation, g_{m1} is the transconductance of stage 1 in Fig. 5, g_{m2} , g_{m3} and g_{m4} are the transconductance of MP5, MP13 and MP6 in Fig. 5, R_1 , R_2 , R_L and C_1 , C_2 , C_L are the equivalent resistance and equivalent capacitance

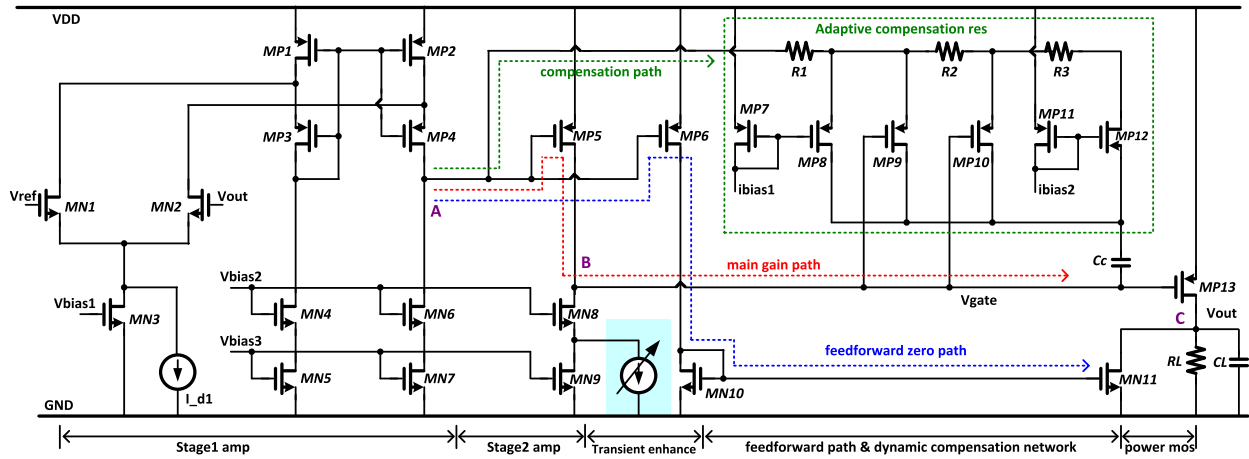


Fig. 5 The total circuit of the proposed LDO.

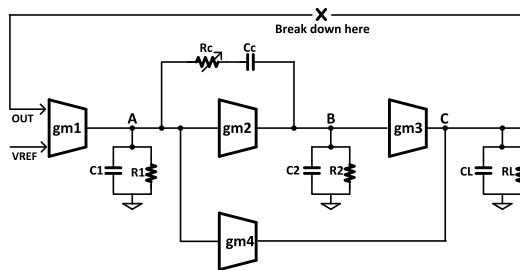


Fig. 6 The small signal model of the proposed LDO.

at node A, B and C in Fig. 5, respectively. R_c and C_c are the compensation resistor and compensation capacitor. In this design, with light load, the position of the main pole is at point C, and the second pole is at point A. But with heavy load, the position of the main pole is at point A, and the second pole is at point C. In order to achieve good stability, the position of the zero position will be designed as close as possible to the position of second pole, thus the second pole and the zero point can cancel each other. So that to achieve good stability at the range of full load.

3. Simulation and analysis

This design is implemented with a 0.5 μm CMOS process. The power supply voltage range is 2.2 V to 5 V, the output voltage range is 1.2 V to 4.5 V. In order to achieve best performance, a 1 μF decouple capacitor will be put at the input and output positions. The overall quiescent power consumption is 12 μA at TT 25 corner by carefully post-simulation. The layout of the chip is shown in Fig. 7, and occupy an area of 0.376 mm^2 .

The PSR curve with and without PSR enhance circuit is shown in Fig. 8. At the typical corner, the output voltage is 2.8 V with the load current 20 mA, the post-simulation result clearly shows the PSR enhance circuit can decrease the ripple at the frequency from 1 to 1 MHz. and improve the PSR -20 dB, -45 dB, -35 dB, -20 dB at the frequency of 1 KHz, 10 KHz, 100 KHz and 1 MHz.

The post-simulation PSR result is shown in Fig. 9, the proposed LDO has the PSR of -83 dB, -68 dB and -52 dB at the frequency of 1 KHz, 100 KHz and 1 MHz. Figure 10

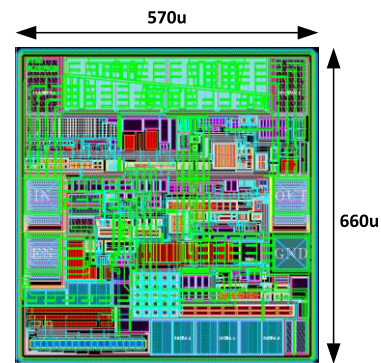


Fig. 7 The layout of the proposed LDO.

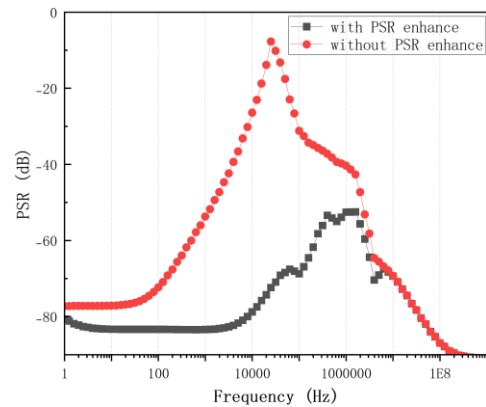


Fig. 8 The post-simulation PSR curve with and without PSR enhance.

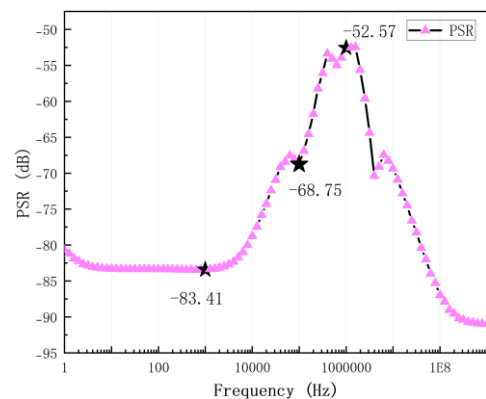


Fig. 9 The typical post-simulation PSR curve.

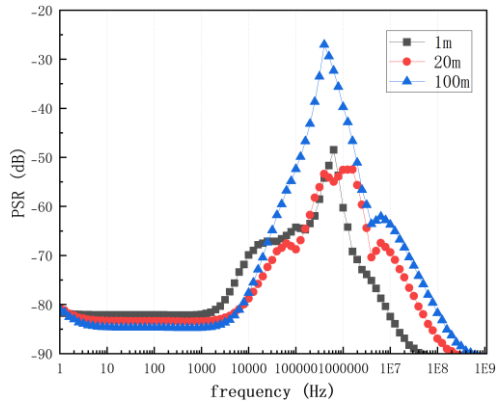


Fig. 10 The post-simulation PSR curve at different load.

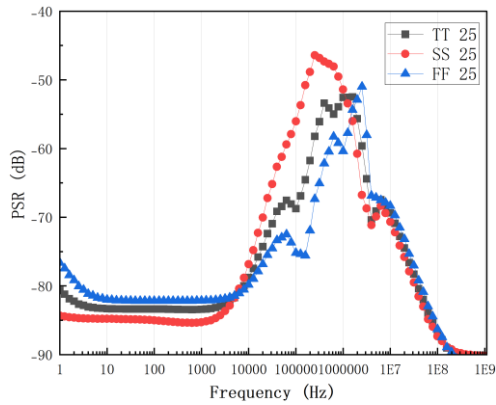


Fig. 11 The post-simulation PSR curve at different corners.

Table I Comparison of the proposed LDO with other works.

	This work	[12]	[22]	[29]	[30]
Tech(μm)	0.5	0.13	0.065	0.18	0.18
Vout(V)	1.2-4.5	1	1.6	1.5	1.8
Cout(pF)	1000	4000	4700	100	260
ILoad(mA)	250	25	25	50	100
IQ(μA)	12	50	71	20	308
PSR(dB) at 1KHz	-83	-80	-61	-95	-75
PSR(dB) at 1MHz	-52	-67	-61	-43	-45
Area(mm^2)	0.376	0.049	0.036	N/A	N/A

shows the PSR curve at different load and Fig. 11 shows the PSR curve at different corner. All the figures verify that the proposed LDO has good PSR at all process.

Table I is the comparison of the proposed LDO with other papers. Compared with the other works, the proposed LDO has a smaller quiescent current, and a bigger load current range, also better PSR performance at 1KHz. But the area of the proposed LDO is larger than other works because of the 0.5 μm technology.

4. Conclusion

This work proposed a high PSR LDO implemented with a 0.5 μm CMOS process. Thanks to the low-frequency PSR enhance technique and the dynamic compensation tech-

nique, this design can achieve ultra-high PSR at low frequency and good stability at full load. The detailed theoretical analysis of PSR enhance and stability analysis is shown in the paper. Besides, the full process and full load post-simulation verifies that this LDO has the advantages of high PSRR, good stability and low quiescent current.

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